

MENG 223 Mechanical Design I Course Project Report
Design Constraints and Requirements Report

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MechaInnovators

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Under supervision of

Prof. Ahmed Sakr

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1-Introduction

Our team, the Mechannowators, has created a human-powered screw press on a small scale that can be used in small workshops and schools. The design priorities are simplicity, low cost, and safety while meeting the needs of a local garage that handles auto body repair and production of small parts. The press will handle tasks such as bearing pressing, straightening of metal components, and light assembly tasks.

The motivation for this project is the need for an affordable, small, and durable press that can work well in space- and resource-scarce settings. By making it manual, we eliminate the expense and complication of motorized systems, making the press accessible to more individuals.



2-Literature Review

2.1-Study of Existing Presses and Their Components

The most ancient press machine developments were in ancient times when crude tools and manual effort were used to shape the materials. The first known presses in history were found by the Greeks in the 1st century AD and used the process of applying the screw press. It was vastly used to cure the pressing of olive oil and grape juice in wine production. The screw press was subsequently further designed by the Romans and continued in application for a stupendous range of uses, including printing books in the 15th century with Johannes Gutenberg's printing press.

Flywheel and crankshaft-driven mechanical press machines were designed with the Industrial Revolution (18th-19th century) in order to increase force and efficiency. Hydraulic presses were invented, initially by Joseph Bramah in 1795. Hydraulic presses became common for manufacturing industries, specifically metal forming, forging, and deep drawing.

In the 20th century, there were also electric and pneumatic presses but with quicker and more accurate control. Modern servo-electric presses today drive computer-aided manufacturing with economy, accuracy, and programmability. Presses made to accommodate growing industry demands, ranging from food processing to the automobile and aeronautical fields, year after year.

Screw presses are mechanical devices widely utilized for use by manufacturing, metalworking, food processing, and waste management industries. They are primarily

One of the largest presses



employed for operations like metal forming, stamping, compression, extrusion, and briquetting. This essay delves into the design, operation, and components of screw presses extensively to provide a comprehensive picture of their function and where they are employed.

1. Power Screw

Power screw is the main component of a screw press that converts rotational energy into linear motion. It applies force to the workpiece required to create the desired deformation or compression.

1- Types of Power Screws:

Lead Screws: Applied in precision operations due to high accuracy.

Ball Screws: Have low friction and high efficiency and are applied in high-speed operations.

Acme Screws: Strong and can sustain heavy loads.

2- Material: Typically made up of high-strength steel or alloy to withstand high stress and wear.

3- Thread Design: The thread profile (square, trapezoidal, or triangular) determines the efficiency and load-carrying ability of the screw.

4- Application: Used in metal forming, forging, and extrusion operations where controlled force must be applied.

2. Bolts

Bolts are used to maintain the structural components of the press stable, offering integrity and stability to work loads.

1- Types of Bolts:

o Hex Bolts: Mostly utilized in general-purpose applications.

o High-Strength Bolts: Utilized where high stress needs to be managed in heavy-duty.

o Anchor Bolts: Fix the press stable to the foundation or floor.

2- Material: Typically manufactured from carbon steel, alloy steel, or stainless steel depending on the load and environmental conditions.

3- Torque Requirements: Proper torque application is required to prevent loosening during operation.

4- Safety Considerations: Bolts need to be periodically checked for wear, corrosion, or fatigue to avert structural collapse.

3. Springs

Screw press springs are energy-absorbing components, supplying the required force in pressing action and enabling smooth operation.

1- Questions on Types of Springs:

- o Compression Springs: Accumulate energy by getting compressed due to load.

- o Torsion Springs: Discharge energy by rotational force.

- o Die Springs: These are specifically used for high-load functions in presses.

2- Material: High-carbon steel or alloy steel is usually used for their manufacture to ensure durability and resilience.

3- Function: Springs serve to provide consistent pressure, dampen vibration, and shield the press from shock loads.

4- Applications: These are employed in stamping, punching, and forming operations for ensuring even force distribution.

4. Belt Drive System

The belt drive system passes the power of the motor to the screw, giving it the rotational motion to perform the press operation.

1- Type of Belts:

- V-Belts: Typically used because of their efficiency as well as high torque capability.

- Timing Belts: Support precise synchronization of the motor and screw.

- Flat Belts: Used in low-power applications.

2- Material: Belts are typically made of rubber, polyurethane, or materials with reinforcement for greater durability.

3- Maintenance: Regular wear, tension, and alignment inspection for optimal performance.

4- Advantages: Economical, vibration-free, and guarantee smooth power transmission.

5. Chassis and Frame

Frame and chassis form the structural backbone of the screw press and are responsible for supporting and stabilizing the machine during operation.

- 1- Material: Constructed of high-strength steel or cast iron to withstand heavy loads and avoid deformation.
- 2- Design Considerations:
- 3- Rigidity: Provides minimum deflection underload.
- 4- Modularity: Easy to assemble, disassemble, and maintain.
- 5- Ergonomics: Designed for operator comfort and safety to operate.
- 6- Applications: The frame must be specifically designed for the specific requirements of the press, such as size, capacity, and working conditions.

6. New Features of Modern Screw Presses

Modern screw presses are designed with enhanced features for better performance, safety, and ease of use:

- 1- Adjustable Speed Ranges: Allow operators to set the speed of the press according to different materials and applications.
- 2- Automation: Programmable logic controller integration to ensure accuracy and repeatability.
- 3- Safety Mechanisms: Incorporate safety stop switches, safety guards, and overload protection to prevent accidents.
- 4- Lubrication Systems: Automatic lubrication systems reduce wear and enhance component life.
- 5- Energy Efficiency: New designs minimize energy consumption while maintaining high levels of performance.

7. Applications of Screw Presses

Screw presses are utility machines employed in many industries:

- 1- Metalworking: Forging, metal extrusion, and metal stamping.
- 2- Food Processing: Juice pressing, oil extraction, and dewatering.
- 3- Waste Management: Compacting waste products for recycling or disposal.

- 4- Pharmaceuticals: Compacting powders and compressing tablets.
- 5- Ceramics and Composites: High-pressure shaping and forming of materials

8. Comparative Analysis of Currently Available Presses

Comparative analysis of current screw presses shows the varying design, capacity, and functionality:

Parameter	Small-Scale Presses	Medium-Scale Presses	Large-Scale Presses
Capacity	Low (up to 50 tons)	Medium (50–200 tons)	High (200+ tons)
Speed	High	Moderate	Low
Applications	Light-duty tasks	General-purpose tasks	Heavy-duty tasks
Automation Level	Manual or semi-automatic	Semi-automatic	Fully automatic
Cost	Low	Moderate	High

9. Challenges and Future Trends

1- Challenges:

- Wear and tear of components due to high stress.
- Maintenance requirements for optimal performance.
- High initial investment for advanced models.

2- Future Trends:

- Integration of IoT for real-time monitoring and predictive maintenance.
- Development of lightweight, high-strength materials for components.
- Increased focus on energy-efficient and eco-friendly designs

2.2-Description of the Main Features and Specifications of a Press

Screw press is a multi-purpose machine used in different industries such as metal working, forging, and sewage treatment. It must be strong, precise, and safe. The following is a detailed description of its main features and specifications:

1. Motor Power

The motor forms the backbone of the screw press. It provides the driving power to run the machine. The motor power determines the pressure applied by the press and at what speed the press runs.

Why it is important: The bigger motor is able to withstand heavier loads and harder work. A workshop's light press, for example, can be fitted with a 3-kW motor, while an industrial press for factories can have motors above 120 kW.

Example: If you are working with thick metal plates, you require a press that has a heavy-duty motor fitted to provide adequate pressure to mold the metal.

2. Speed Range

Speed range is the rate at which the screw press can move. Most of the screw presses come with adjustable speed, and you should be able to since different jobs and materials require different speeds.

Why it matters: For accurate work, you would need to work slowly in order to be accurate.

In order to work quickly, you would work at a higher speed in order to save time. Example:

While stamping a design on thin metal, you would need to work at a slow speed so that you do not damage the metal. But if you are working on numerous parts, you would need to work at a fast speed so that you work effectively.

3. Bed Length

Bed length refers to width of work space upon which you mount the material or workpiece. It determines the pieces to which you are able to work, either big or small.

Why is it important: The big bed allows you to work with longer or bigger materials, and a small bed takes up minimal space and is suitable for extremely small jobs.

Example: One can use a small press with bed length of 300 mm for small-sized metal pieces, while a large press with bed length of 1800 mm and more can be used for large work size, i.e., metal plate.

4. Tool Rest

Tool rest is a stiff base upon which you place tools or dies. It keeps everything secured well and makes it work.

Why it matters: A stiff tool rest prevents shaking and makes the workpiece get formed as it is supposed to. In the absence of a proper tool rest, the material would move around, resulting in errors or poor output.

Example: During forging a metal part, the tool rest maintains the die steady so that the metal gets formed just as needed.

5. Ergonomics Design

Ergonomics is all about making the machine comfortable and easy to use for the operator. Modern screw presses are fatigue-reducing in design and reducing the work from being tiring.

Why it matters: When operators do not have to work under stress, they can work longer and more effectively without fatigue.

Features:

Controls that are easy to operate so that operators do not need to stretch and bend uncomfortably.

Adjustable controls to suit the operator's own preference.

Simple interfaces, e.g., touch screens or buttons, to make it easy to operate the machine.

6. Safety Features

Safety is probably the most important feature of any industrial machine. Screw presses have several safety features to protect the operator and prevent accidents.

Why it matters: Heavy machinery is dangerous to drive, and that's why safety features are necessary to avoid harm and damage to the equipment.

Safety Key Features:

Emergency Stop Buttons: These enable the driver to abruptly shut the machine in case something happens.

Guards: These are placed over moving parts so that the driver will not unknowingly touch them.

Overload Protection: It stops the machine from running when pressure is too high, and it can ruin the press or workpiece.

Other Specifications

In addition to the major specifications, screw presses have other technical specifications that determine how they operate:

Force Capacity (Tonnage):

The maximum force of the press, measured in tons.

Why it matters: To make heavy-duty applications, you must have a press with a high force capability to mold thick or tough material.

Ram Stroke:

This is the distance the ram (the moving part) is able to travel.

Why it matters: The more the stroke, the thicker or taller materials you can process.

Screw Pitch:

This is the gap between the threads of the screw. It influences the precision of the press and the pressure that can be achieved.

Why it matters: The lower pitch (closely spaced threads) has more precision, but larger pitch (wider spaced) has faster motion.

Automation:

Most modern screw presses have some form of automation, either programmable controls or sensors, to increase efficiency in the process.

Why it is so: Automation speeds up the machine and reduces the work required, hence the process is efficient and less error-prone.

Applications of Screw Presses

Screw presses have applications in different industries for different uses, i.e.,

Forging: Shaping metal pieces, e.g., tools or gears.

Stamping and Forming: Producing patterns or shapes on materials such as metal sheets.

Dewatering: Driving out water from materials such as from food processing or wastewater.

These machines are very resilient, precise, and flexible for operating a wide range of different types of jobs, so they are very applicable in industry.

For our project, our group has opted to concentrate on the conceptual and design of a small-scale manually driven screw press. This is in line with the aim of developing a functional, economical, and easy-to-use machine appropriate for settings like small workshops or learning environments, where simplicity, strength, and restricted application of force are of essence. The following sections will outline the precise requirements that this kind of press requires.

2.3-Design Requirements of a Small-Scale Screw Press

A screw press is a type of mechanical press that applies a helical screw mechanism to generate high-pressure force to work various materials. A small-scale screw press design must adhere to certain constraints and requirements to be efficient, cost-effective, durable, and easy to use.

Critical Design Constraints and Requirements

1. Budget Constraints

The total material and manufacturing costs must be within budget constraints. Some commercially employed machines are as follows:

-Mini Expeller Screw Press Oil Maker Machine: \$56,000

-Mild Steel Commercial Expeller Screw Press Machine: \$1,600,000

-Automatic Dewatering Screw Press Machine: \$280,000

The structure must be designed to minimize costs in material selection and manufacturing operations without compromising on quality.

2. Space Requirements

The press must be fit into a previously allocated space, so small-sized designs are best. Typical dimensions to refer to are:

- Mini Expeller Screw Press Oil Maker Machine: 350x160x360mm

- YZS-68 Mini Oil Expeller: 900x390x750mm

Modular parts for easy installation and relocation in small spaces should be considered in the design.

3. Power Supply Considerations

The press must be designed to work with standard power supply requirements:

- Mini Expeller Screw Press Oil Maker Machine: 220V

- Mild Steel Commercial Expeller Screw Press Machine: 415V

Energy-efficient motors must be employed for reduced operational expenses and increased sustainability.

4. Load Capacity

The press should be able to exert a force strong enough to perform the desired processing. Key capacity specifications are:

- Mini Expeller Screw Press Oil Maker Machine: 1200W

- Mild Steel Commercial Expeller Screw Press Machine: 56 kW

- YZS-68 Mini Oil Expeller: 5.5 kW

Screw mechanism and motor should be strong enough to resist the required load without excessive wear or breakdown.

5. Safety Features

For user safety, the design should be incorporated:

- Emergency stop features for instant shutdown.

- Overload protection against burnout and structural failure.
- Good shielding around moving parts to prevent accidental injury.

6. Durability and Material Selection

Structural components must be made of rugged materials:

- Cast Iron Bed: In large-scale applications as it provides vibration resistance.
- Stainless Steel Body: Provides corrosion resistance and durability.
- Mild Steel or Stainless-Steel Frame: Supports the structure while maintaining the weight minimum.
- High-Tensile Alloy Screws: Reduces wear and increases operating life.

7. Ease of Maintenance

The design should be repairable and maintainable with ease by the incorporation of:

- Modular assembly to easily disassemble and reassemble.
- Lubrication points to reduce friction and wear by making them accessible.
- Replaceable elements such as bearings and seals to ensure maximum machine lifespan.

Additional Design Considerations

1. Efficiency and Performance Optimization

- Gear Reduction System: Maximizes torque output while reducing power consumption.
- Variable Speed Control: Allows users to control processing speed based on material requirements.
- Heat Dissipation System: Prevents overheating during extended operation.

2. Noise and Vibration Control

- Anti-vibration Mounts: Reduce operational noise and improve stability.
- Sound Insulation Panels: Minimize environmental noise pollution at the workplace.

-Precision Engineering: Avoids misalignment and useless friction, maximizes efficiency.

3. Environmental Conditions

-Waste Management System: Grapes and processes left material with utmost efficiency.

-Energy-Efficient Motors: Save power consumption and carbon emission.

-Biodegradable Lubricants: Have minimal environmental effect compared to conventional oils.

2.4-Customer Requirement-Based Design Constraints Definition

Small automotive body repair and manufacturing garage. They are in need of a screw press for small to medium-sized manufacture workshops for production. They aim to create a cost-saving, space-efficient, and durable machine according to workshop standards but with safety and efficiency assurances.

Customer Profile:

- Industry: Small and medium-sized manufacture workshops
- Budget: Limited budget, thus cost-saving design solutions are required
- Space Availability: Small workshops with limited floor area
- Power Supply: Routine workshop power supply
- User Experience: Routine technical skills operators

Decision-Making Process:

1. Budget Consideration:

- The use of mild steel as the frame instead of expensive alloys.
- They limit the use of electronic controls, applying manual or semi-automatic operation.

2. Space Management:

- The press is built vertically in an effort to save floor space.
- Has a folding worktop to accommodate different sizes of workshops.

3. Power Supply Restraints

- Motor selection is designed for minimal power consumption with adequate torque.
- All other hand tools are designed for areas with an unreliable supply of electricity.

4. Safety Features:

- Find an uninterrupted safety guard against accidental hand cutting.
- Provide an emergency stop switch in easy reach.
- The belt drive is housed in an entanglement-preventing casing.

5. Convenience of Use Considerations:

- Press handle is ergonomically designed for least effort involved.
- Single button start/stop facility is convenient to operate.

6. Durability & Long-Term Reliability:

- Screw is hardened steel with corrosion-resistant finish.
- Frame is over-designed to withstand repeated heavy loading.

Benefits of the design:

- It can be applied in metal part straightening, pressing bearings, and assembling parts.
- It is fitted in a working workshop where space is restricted.
- Mechanics will use it with minimal training.
- The robust safety feature discourages accidents in a busy workshop.

Design Specification

From the requirements set for a low-cost, light, and hard-wearing screw press that would be used in a neighborhood garage performing auto bodywork and manufacturing small-sized parts, the following key requirements have been set for our screw press manual:

Motor Power (or Manual Operation Input)

- **Requirement:** The press is operated manually, with the cost and mechanical complexity of a motor minimized.
- **Operator Input:** The operation is easy with an estimated uniform operator force of 500 Newtons (N) on the handle.
- **Leverage:** The principal operating handle is constructed with a length of approximately 250 mm (center of rotation to grip point) to provide sufficient mechanical advantage for the operations.

Speed Range (Ram Travel Rate)

- **Specification:** As it is a hand-operated press, ram travel rate will be governed by the speed at which the operator operates. Design calls for controlled force and precision but not speed.
- **Screw Pitch & Travel:** The operating screw will have a square thread with a 3 mm pitch. It

provides a 3 mm vertical ram travel per complete 360-degree turn of operating handle.

Bed Length (and Working Area)

- **Customer Requirement:** Floor area is limited locally but flexibility must be accommodated.
- **Fixed Bed Area:** The main fixed bed (lower support surface) is about 750 mm (length) x 360 mm (width) working area. This is suitable for standard tasks like pressing bearings and assembly of small components.
- **Daylight Vertical:** The ram (fully retracted) to bed maximum vertical distance (daylight) to be approximately 565 mm to accommodate different heights of workpieces.

Tool Rest / Platen

- **Upper Platen (Ram Face):** To touch top of moving ram (upper platen) surface is a Ø230 mm steel alloy disc-shaped plate to bear pressing force and wear.
- **Lower Bed/Support:** The lower bed is a robust, flat tabletop integral to the frame to hold tooling such as V-blocks, press plates, or special fixtures in direct support. It consists of steel alloy

Ergonomic and Safety Considerations

- **Ergonomics**
 - **Handle Position:** The handle for operating is placed at a nominal height of 1000 mm above the ground when the press is placed on a typical workbench so that it can be easily operated by an ordinary individual.
 - **no Handle Grip:** The handle incorporates a grip to reduce friction on the hands of the operator and allow smooth, prolonged effort.
- **Safety:**
 - o **Frame Stability:** The frame is constructed with a wide base to ensure stability during running and prevent tilting.
 - o **Clearance & Pinch Points:** The design is constructed to reduce pinch points, and the primary pressing area remains open for easy sight.
 - o **Controlled Operation:** That the screw mechanism is manual naturally guarantees controlled application and withdrawal of force. There is no stored energy, which can lead to random motion.

Target Force Capacity

- **-Specification:** The screw press can attain a nominal maximum pressing force of approximately 1.228 kN.
- **Reasoning:** The force capability for this is deemed sufficient for the work for which the local garage is designed, i.e., squeezing small and medium bearings, bushings, light-duty

metal part straightening, and assembly work, according to the leverage achievable with the handle and chosen screw mechanism.

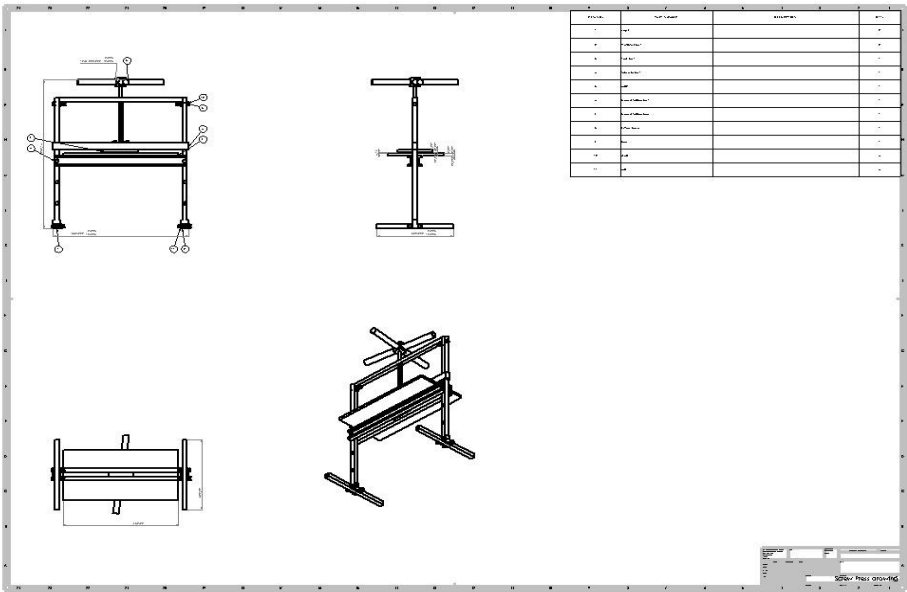
3-Conceptual Design

Overview of Conceptual Design:

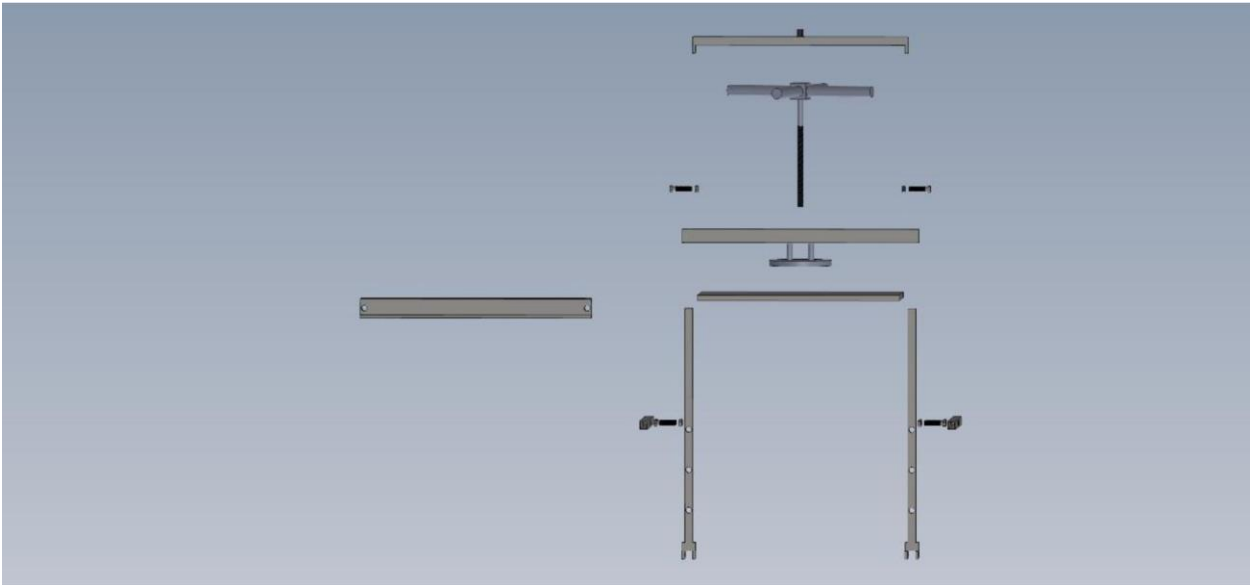
The conceptual design of the screw press project is a heavy-duty, manually operated U-frame screw press, specifically tailored to the needs of a local garage. The structure of the press consists of welded steel alloy components to provide maximum strength-to-cost ratio. The press is powered through a central square power screw. The screw is powered by a manually driven, ergonomically shaped handle that advances a guided ram in a vertical motion to exert compressive force. The design is simplified, toughened, and made secure and provides an easy solution to such tasks as bearing insertion and straightening small parts, directly addressing the customer requirements for a small machine that can be handled with ease.

3D CAD Model:

Following are the 3D CAD views of the conceptual screw press conceptualized using SolidWorks software. These views reveal the overall assembly and major components. Front, Side, Top, and isometric view for Screw Press Assembly with Bill Of Material and balloons:



Exploded View for the Screw Press Assembly:



Mass Properties for the Screw Press Assembly:

Mass properties of Screw Press

Configuration: Default

Coordinate system: -- default --

Mass = 96224.89 grams

Volume = 12401775.22 cubic millimeters

Surface area = 2361080.67 square millimeters

Center of mass: (millimeters)

X = 1047.97

Y = 1339.01

Z = 1276.68

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

I_x = (0.00, 1.00, 0.00)P_x = 5591212329.40I_y = (-1.00, 0.00, 0.00)P_y = 16263260470.16I_z = (0.00, 0.00, 1.00)P_z = 20512343162.04

Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

L_{xx} = 16263260470.16L_{xy} = -1918.52L_{xz} = 125.11L_{yx} = -1918.52L_{yy} = 5591407215.01L_{yz} = 53924722.71L_{zx} = 125.11L_{zy} = 53924722.71L_{zz} = 20512148276.43

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

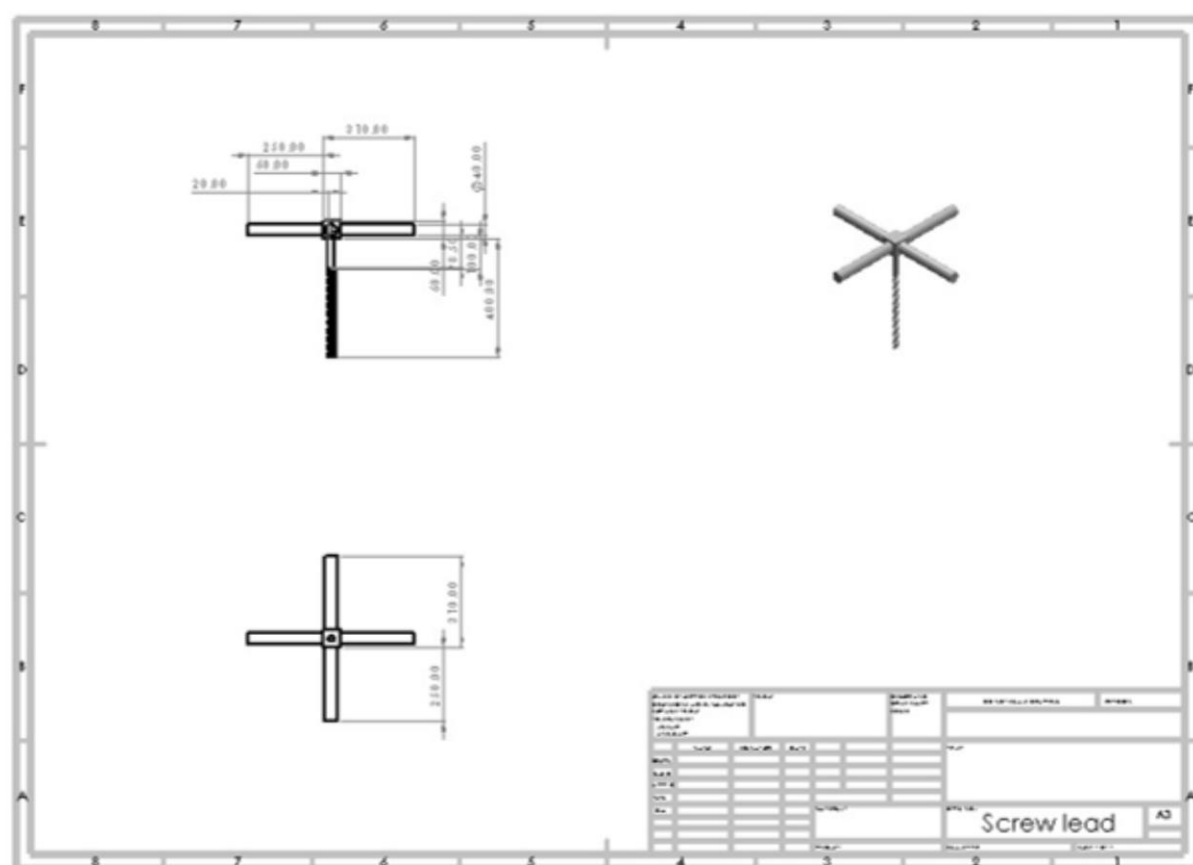
I_{xx} = 345625905614.04I_{xy} = 135026284248.12I_{xz} = 128741017018.49I_{yx} = 135026284248.12I_{yy} = 268106737297.52I_{yz} = 164548336806.08I_{zx} = 128741017018.49I_{zy} = 164548336806.08I_{zz} = 298715231211.78**Bill of Material:**

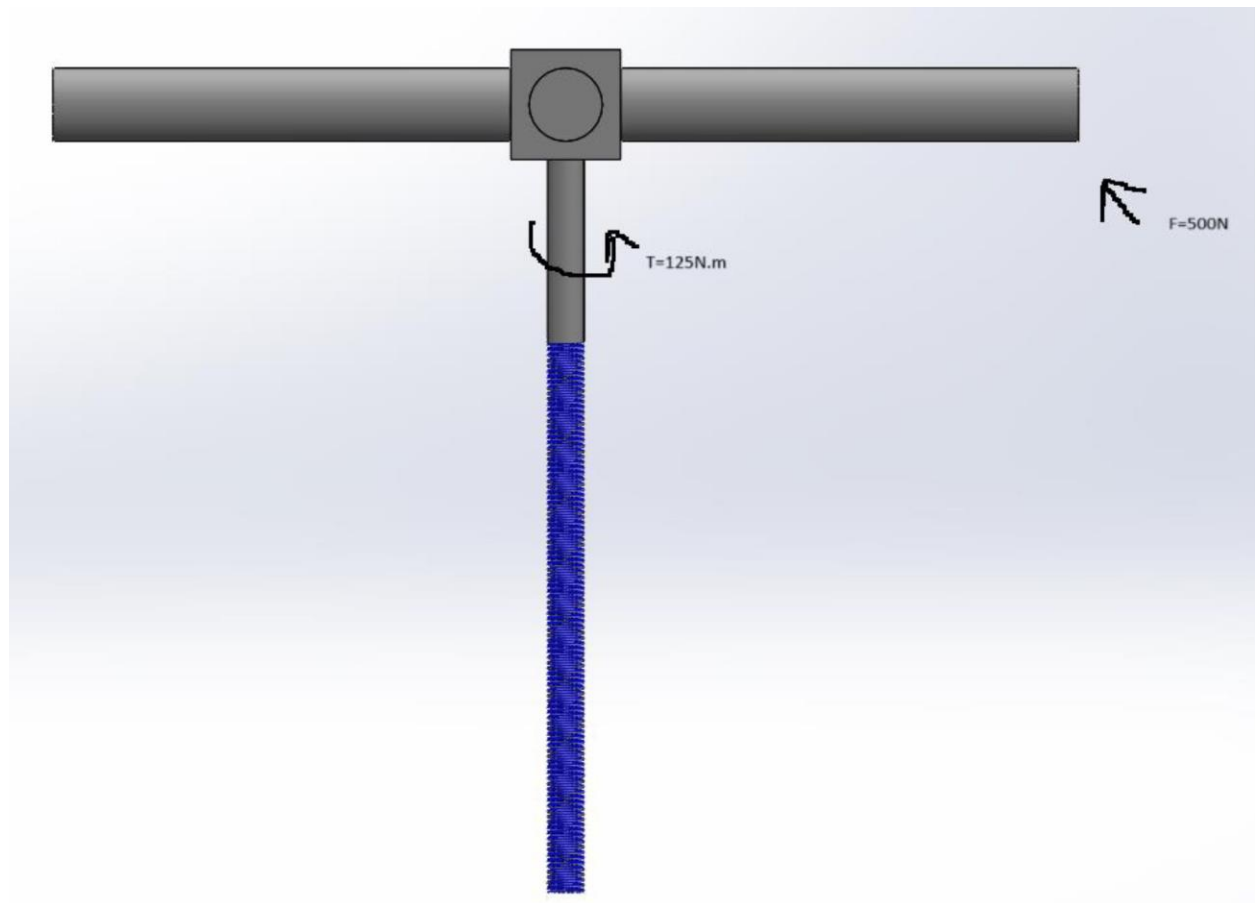
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Legs	Costume	2
2	Vertical Bar	Costume	2
3	Top Bar	Costume	1
4	Middle Bar	Costume	1
5	nut2	Standard	1
6	Compression Bar	Costume	1
7	Compression Bed	Costume	1
8	Screw lead	Costume	1
9	Bed	Costume	1
10	bolt	Standard	6
11	nut	Standard	6

Detailed Drawings & Analysis of Key Components:

This section provides detailed drawings and preliminary engineering considerations for key custom-designed components of the screw press. For each component, a SolidWorks drawing, a Free Body Diagram (FBD) illustrating expected operational forces.

The Handle and the Power Screw Drawing and the free body diagram with the force and torque acting on them

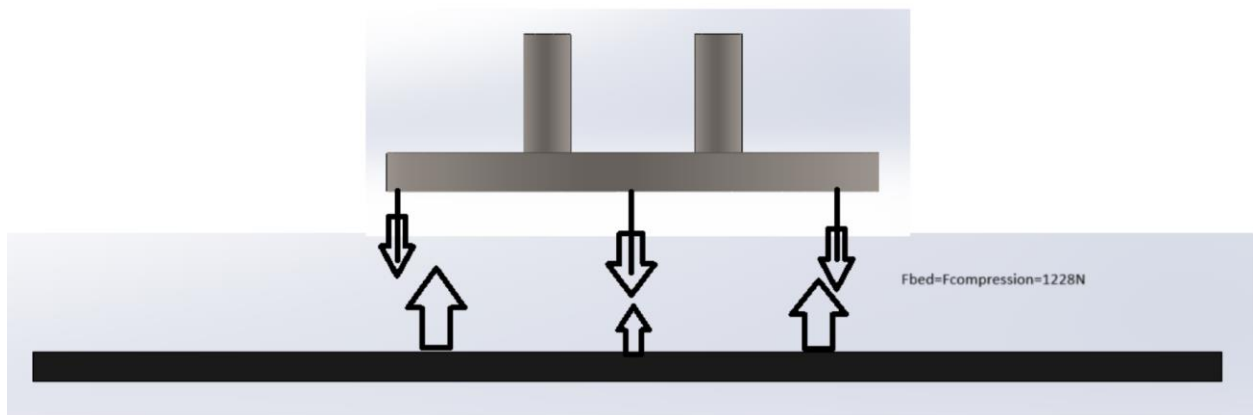
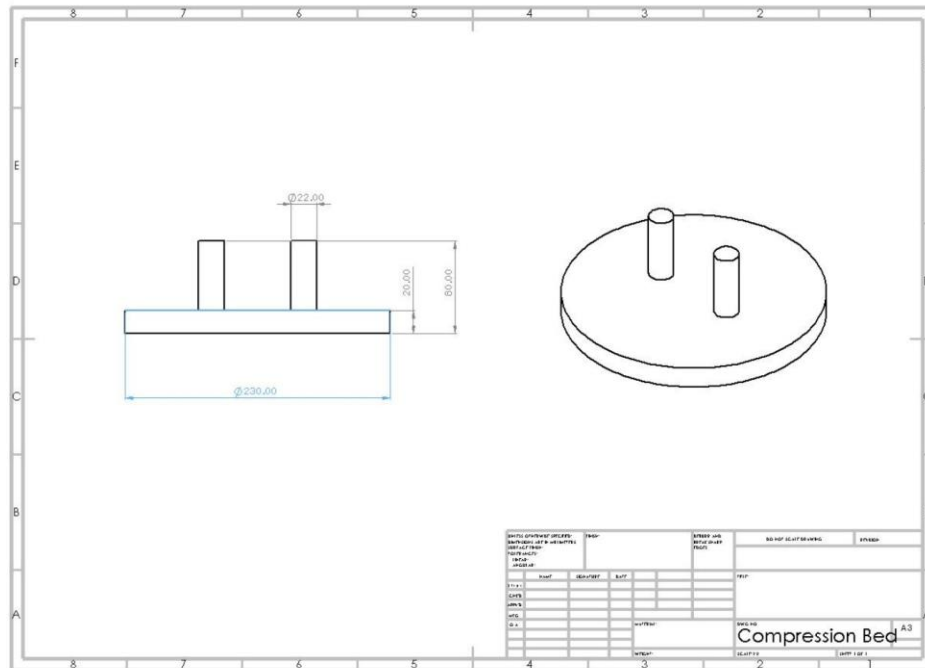




Material Selection: Steel Alloy

Expected Types of Stress: Torsional Shear Stress, Compressive Axial Stress

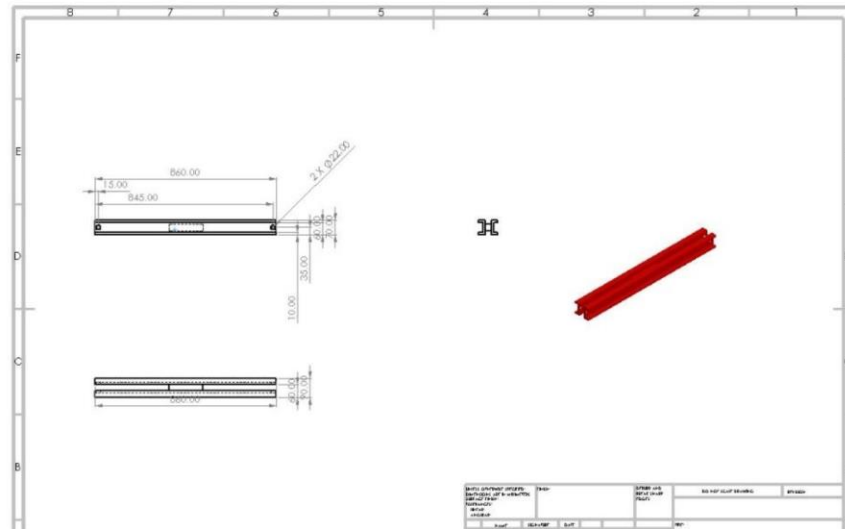
The Compression Bed Drawing and free body diagram with the resultant reactions on it



Material Selection: Steel Alloy

Expected Types of Stress: Compressive Axial Stress

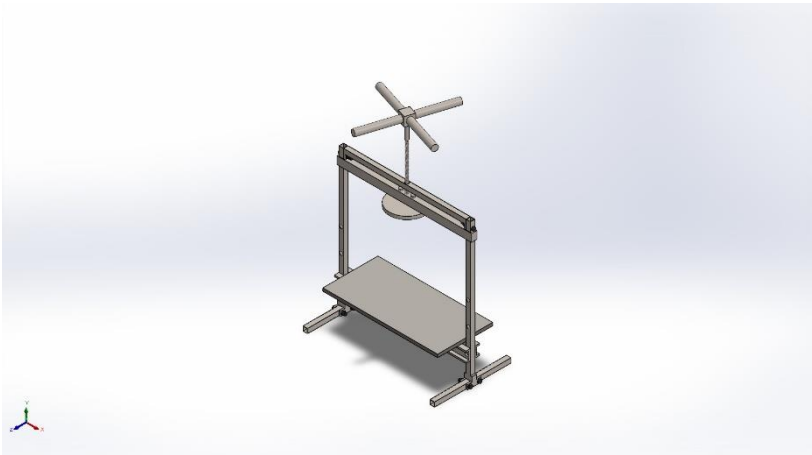
The Middle part of the frame drawing



Material Selection: Steel Alloy

Expected Types of Stress: Compressive Axial Stress, Bending Moment

4-Stress Analysis



Description

The calculation were done to insure the safety and efficiently of the screw press:

Capacity: Estimated Based on the screw dimensions, pitch, material, bed to determine the size of the material for the press.
Power requirement: Calculated from the Capacity, shaft length, gravity to and material factor to determine the force.
Shaft Diameter and stress: analyzing bending and torsional moments, factor of safety and allowable shear stress.
Friction and forces: understanding load transfer and mechanical efficiency, thread design and shaft strength

Simulation of Screw Press

Date: Tuesday, May 13, 2025
Designer: Solidworks
Study name: Static 1
Analysis type: Static

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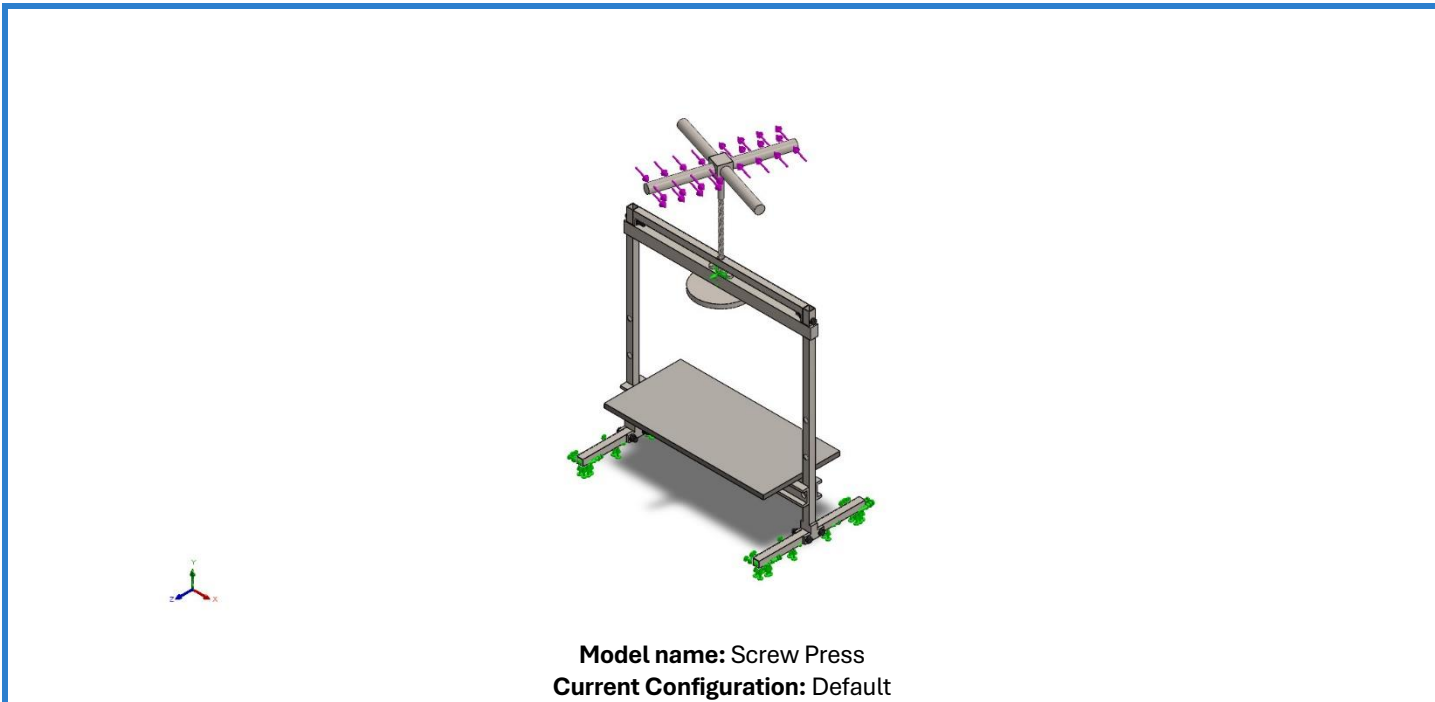
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Assumptions

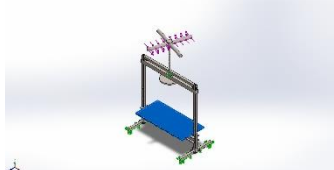
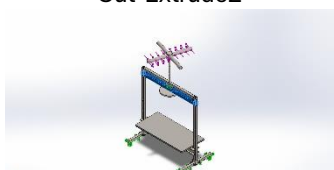
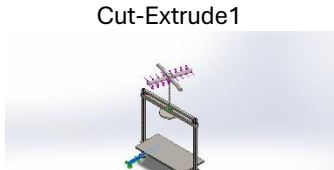
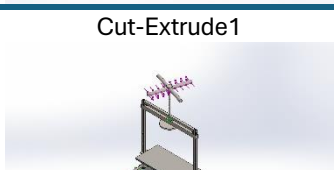
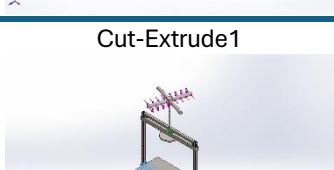
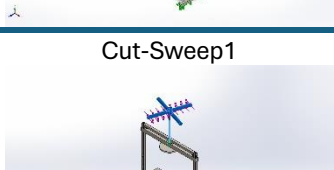
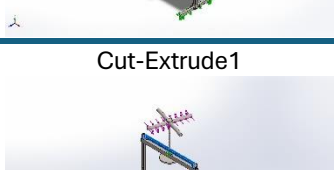
The following assumptions were made during the development of this conceptual design to simplify analysis and focus on the primary design elements.

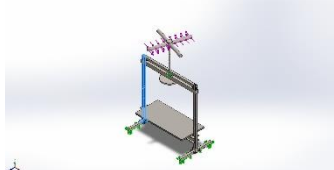
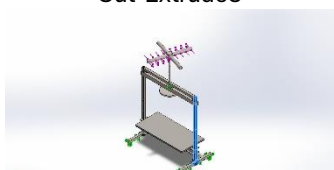
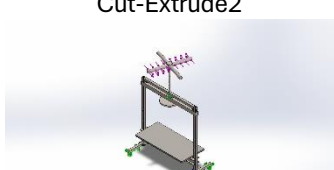
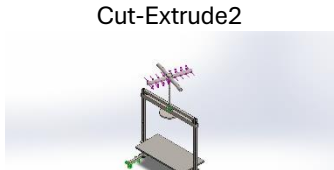
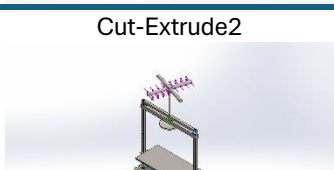
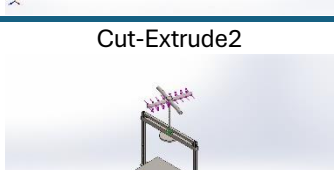
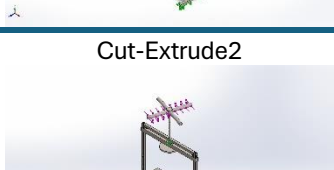
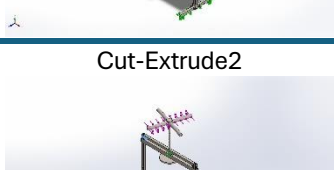
- Material properties are based on nominal handbook values for the selected materials.
- Connections between frame members (e.g., welds, bolts) are assumed to be sufficiently rigid for conceptual load transfer.
- "The coefficient of friction for the power screw mechanism is estimated to be 0.15.
- Operator input force is assumed to be applied steadily and at the designated points on the handle.
- Detailed analysis of stress concentrations at features like fillets, holes, or thread roots is reserved for a later design stage.
- Manufacturing tolerances, fits, and surface finishes are not specified in detail at this conceptual stage.
- The effects of long-term wear and fatigue are not quantitatively assessed at this conceptual level.

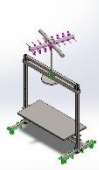
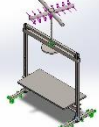
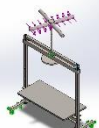
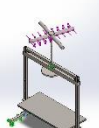
Model Information



Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified

Boss-Extrude1 	Solid Body	Mass:42.39 kg Volume:0.0054 m³ Density:7,850 kg/m³ Weight:415.422 N	C:\Users\Moamen\Downloads\Bed.SLDPRT May 13 16:08:26 2025
Cut-Extrude2 	Solid Body	Mass:11.2111 kg Volume:0.00145599 m³ Density:7,700 kg/m³ Weight:109.869 N	C:\Users\Moamen\Downloads\Compression Bar.SLDPRT May 13 14:39:48 2025
Boss-Extrude2 	Solid Body	Mass:6.74957 kg Volume:0.000876567 m³ Density:7,700 kg/m³ Weight:66.1458 N	C:\Users\Moamen\Downloads\Compression Bed.SLDPRT May 13 15:37:40 2025
Cut-Extrude1 	Solid Body	Mass:0.846725 kg Volume:0.000109964 m³ Density:7,700 kg/m³ Weight:8.2979 N	C:\Users\Moamen\Downloads\Legs.SLDPRT May 13 14:35:34 2025
Cut-Extrude1 	Solid Body	Mass:0.846725 kg Volume:0.000109964 m³ Density:7,700 kg/m³ Weight:8.2979 N	C:\Users\Moamen\Downloads\Legs.SLDPRT May 13 14:35:34 2025
Cut-Extrude1 	Solid Body	Mass:15.5601 kg Volume:0.00202079 m³ Density:7,700 kg/m³ Weight:152.489 N	C:\Users\Moamen\Downloads\Middle Bar.SLDPRT May 13 14:34:03 2025
Cut-Sweep1 	Solid Body	Mass:12.2059 kg Volume:0.00158518 m³ Density:7,700 kg/m³ Weight:119.618 N	C:\Users\Moamen\Downloads\Screw lead.SLDPRT May 13 22:45:48 2025
Cut-Extrude1 	Solid Body	Mass:1.46754 kg Volume:0.00019059 m³ Density:7,700 kg/m³ Weight:14.3819 N	C:\Users\Moamen\Downloads\Top Bar.SLDPRT May 13 14:34:37 2025

Cut-Extrude3 	Solid Body	Mass:1.98538 kg Volume:0.000257842 m³ Density:7,700 kg/m³ Weight:19.4567 N	C:\Users\Moamen\Downloads\Vertical Bar.SLDPRT May 13 14:33:22 2025
Cut-Extrude3 	Solid Body	Mass:1.98538 kg Volume:0.000257842 m³ Density:7,700 kg/m³ Weight:19.4567 N	C:\Users\Moamen\Downloads\Vertical Bar.SLDPRT May 13 14:33:22 2025
Cut-Extrude2 	Solid Body	Mass:0.142118 kg Volume:2.00165e-05 m³ Density:7,100.03 kg/m³ Weight:1.39275 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025
Cut-Extrude2 	Solid Body	Mass:0.142117 kg Volume:2.00165e-05 m³ Density:7,100 kg/m³ Weight:1.39275 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025
Cut-Extrude2 	Solid Body	Mass:0.142117 kg Volume:2.00165e-05 m³ Density:7,099.98 kg/m³ Weight:1.39274 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025
Cut-Extrude2 	Solid Body	Mass:0.142117 kg Volume:2.00165e-05 m³ Density:7,099.99 kg/m³ Weight:1.39275 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025
Cut-Extrude2 	Solid Body	Mass:0.142117 kg Volume:2.00165e-05 m³ Density:7,100.01 kg/m³ Weight:1.39275 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025
Cut-Extrude2 	Solid Body	Mass:0.142117 kg Volume:2.00165e-05 m³ Density:7,099.99 kg/m³ Weight:1.39275 N	C:\Users\Moamen\Downloads\bolt.SLDPRT May 13 22:49:14 2025

Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Thread1 	Solid Body	Mass:0.0132596 kg Volume:1.86749e-06 m³ Density:7,100.21 kg/m³ Weight:0.129944 N	C:\Users\Moamen\Download s\nut.SLDPRT May 13 22:49:18 2025
Boss-Extrude3 	Solid Body	Mass:0.0441778 kg Volume:5.73714e-06 m³ Density:7,700.32 kg/m³ Weight:0.432942 N	C:\Users\Moamen\Downloa ds\nut2.SLDPRT May 13 15:37:18 2025

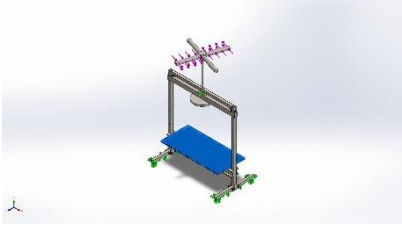
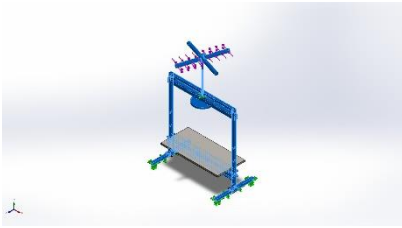
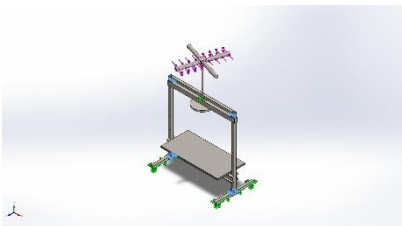
Study Properties

Study name	Static 1
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\Moamen\Downloads)

Units

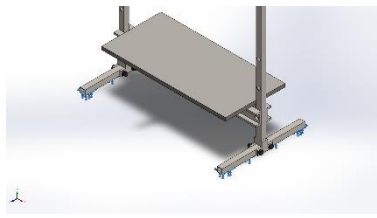
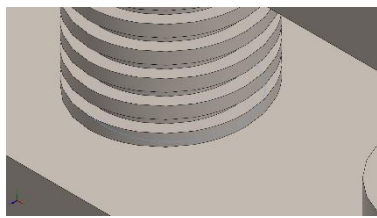
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

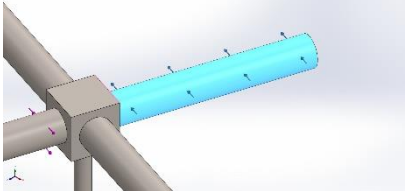
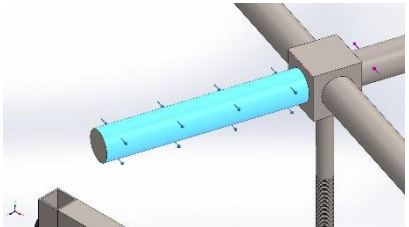
Material Properties

Model Reference	Properties	Components
	Name: AISI 1045 Steel, cold drawn Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 5.3e+08 N/m ² Tensile strength: 6.25e+08 N/m ² Elastic modulus: 2.05e+11 N/m ² Poisson's ratio: 0.29 Mass density: 7,850 kg/m ³ Shear modulus: 8e+10 N/m ² Thermal expansion coefficient: 1.15e-05 /Kelvin	SolidBody 1(Boss-Extrude1)(Bed-1)
Curve Data:N/A		
	Name: Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 6.20422e+08 N/m ² Tensile strength: 7.23826e+08 N/m ² Elastic modulus: 2.1e+11 N/m ² Poisson's ratio: 0.28 Mass density: 7,700 kg/m ³ Shear modulus: 7.9e+10 N/m ² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Cut-Extrude2)(Compression Bar-1), SolidBody 1(Boss-Extrude2)(Compression Bed-1), SolidBody 1(Cut-Extrude1)(Legs-1), SolidBody 1(Cut-Extrude1)(Legs-2), SolidBody 1(Cut-Extrude1)(Middle Bar-1), SolidBody 1(Cut-Sweep1)(Screw lead-1), SolidBody 1(Cut-Extrude1)(Top Bar-1), SolidBody 1(Cut-Extrude3)(Vertical Bar-1), SolidBody 1(Cut-Extrude3)(Vertical Bar-2), SolidBody 1(Boss-Extrude3)(nut2-1)
Curve Data:N/A		
	Name: Ductile Iron Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 5.51485e+08 N/m ² Tensile strength: 8.61695e+08 N/m ² Elastic modulus: 1.2e+11 N/m ² Poisson's ratio: 0.31 Mass density: 7,100 kg/m ³ Shear modulus: 7.7e+10 N/m ² Thermal expansion 1.1e-05 /Kelvin	SolidBody 1(Cut-Extrude2)(bolt-1), SolidBody 1(Cut-Extrude2)(bolt-2), SolidBody 1(Cut-Extrude2)(bolt-3), SolidBody 1(Cut-Extrude2)(bolt-4), SolidBody 1(Cut-Extrude2)(bolt-5), SolidBody 1(Cut-

	coefficient:	Extrude2)(bolt-6), SolidBody 1(Thread1)(nut-1), SolidBody 1(Thread1)(nut-2), SolidBody 1(Thread1)(nut-3), SolidBody 1(Thread1)(nut-4), SolidBody 1(Thread1)(nut-5), SolidBody 1(Thread1)(nut-6)
Curve Data:N/A		

Loads and Fixtures

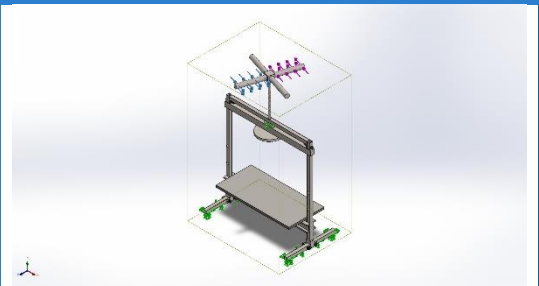
Fixture name	Fixture Image	Fixture Details		
Fixed-1		Entities: 2 face(s) Type: Fixed Geometry		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-0.950753	-2.2714	-0.867553	2.61072
Reaction Moment(N.m)	0	0	0	0
Fixed-2		Entities: 1 edge(s) Type: Fixed Geometry		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	0.684814	2.36401	1.1377	2.71143
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s) Reference: Edge< 1 > Type: Apply force Values: ---, ---, 500 N
Force-2		Entities: 1 face(s) Reference: Edge< 1 > Type: Apply force Values: ---, ---, 500 N

Connector Definitions

No Data

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		Type: Bonded Components: 1 component(s) Options: Independent mesh

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	61.4632 mm
Minimum element size	6.05977 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

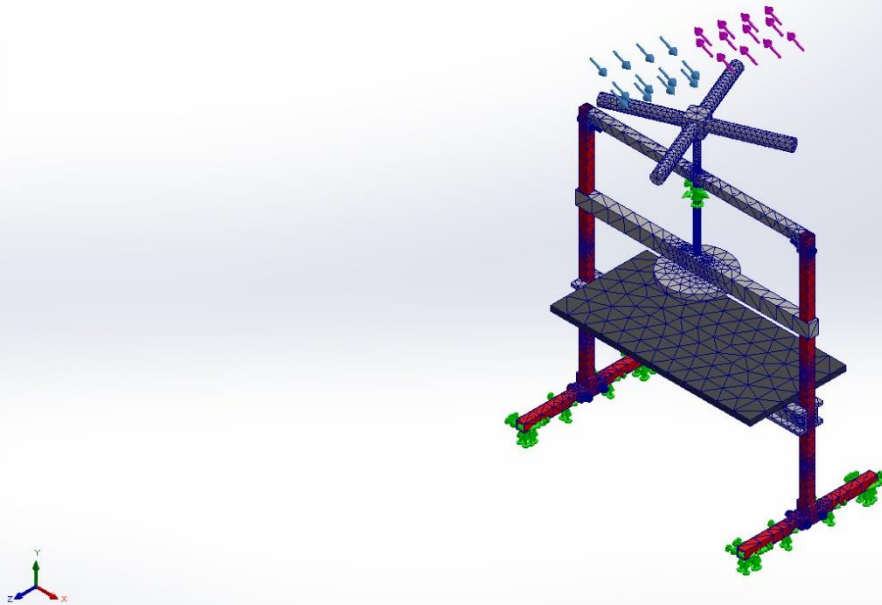
Mesh information - Details

Total Nodes	114664
Total Elements	60182
Maximum Aspect Ratio	1.0974e+05
% of elements with Aspect Ratio < 3	30.6
Percentage of elements with Aspect Ratio > 10	19.4
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:01:08
Computer name:	MOAMEN-LOQ

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-

Model name: Screw Press
Study name: Static 1(-Default-)
Plot type: Mesh Quality1



Screw Press-Static 1-Quality-Quality1

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.265991	0.0926094	0.270264	0.390346

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.0916405	11.915	11.8527	16.8067

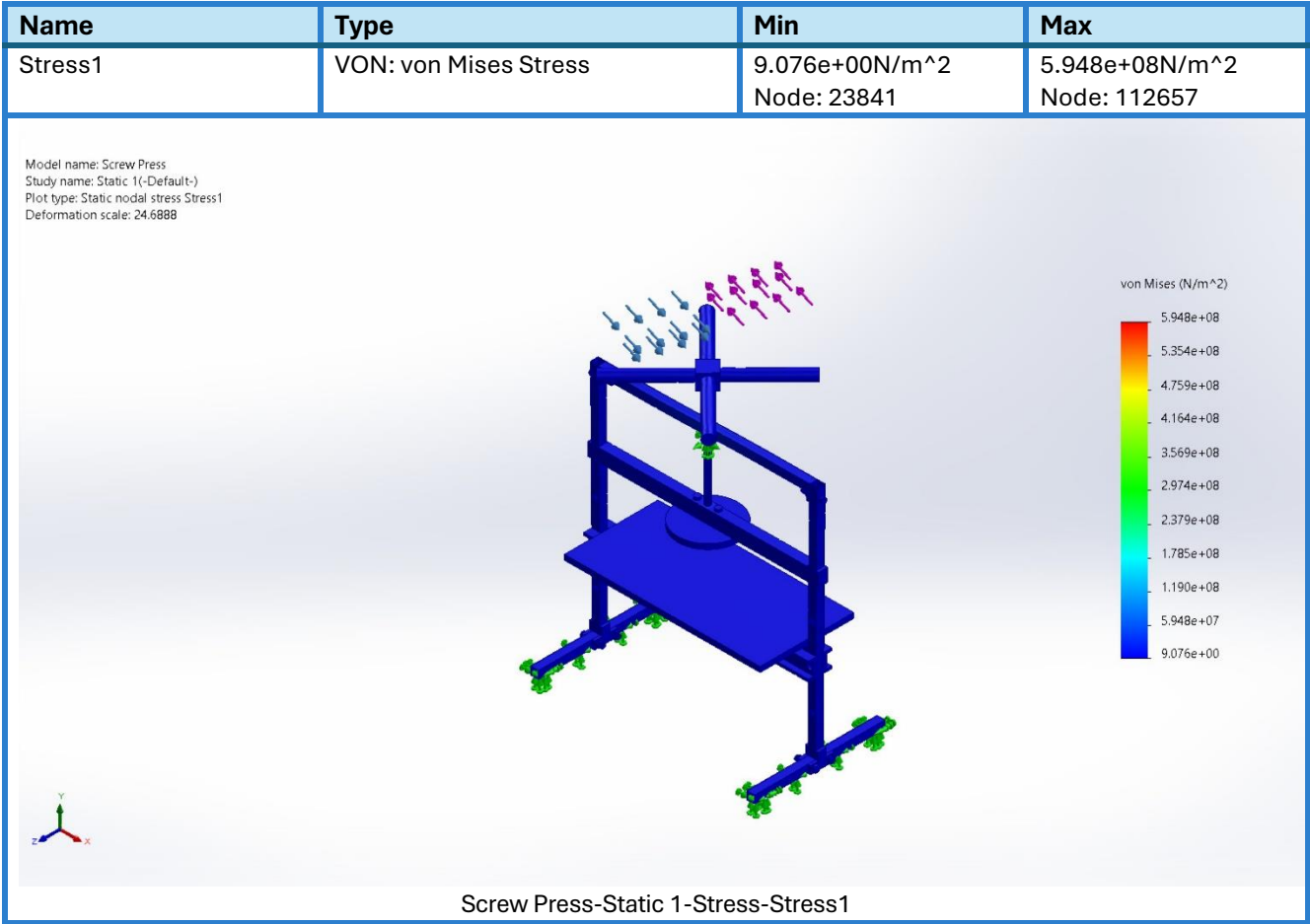
Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

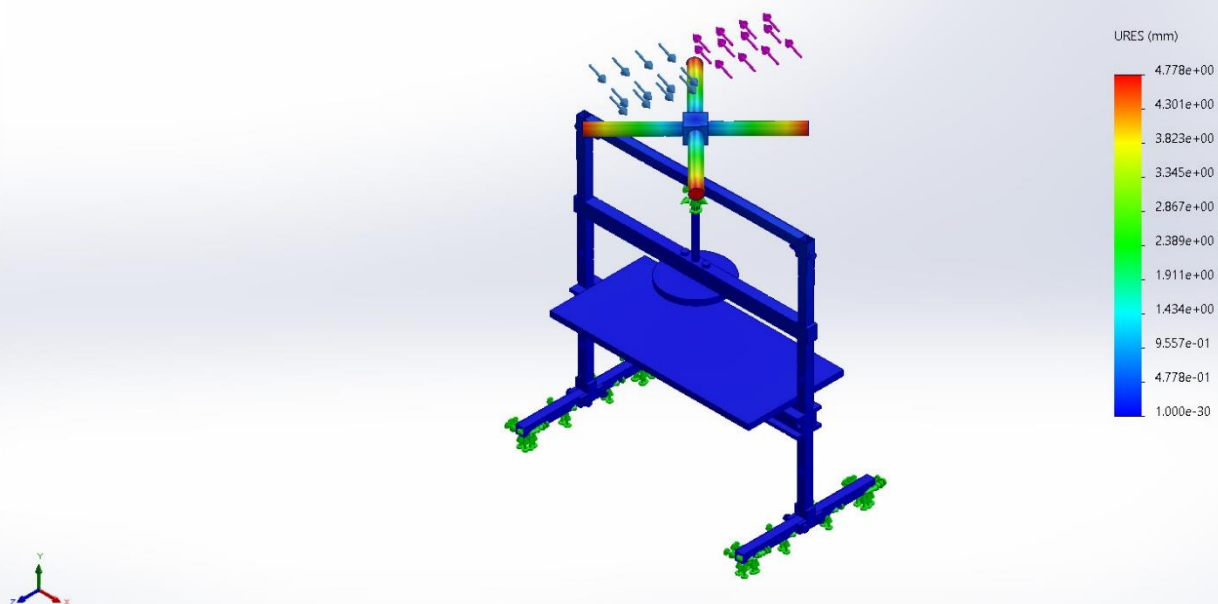
No Data

Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 5206	4.778e+00mm Node: 12982

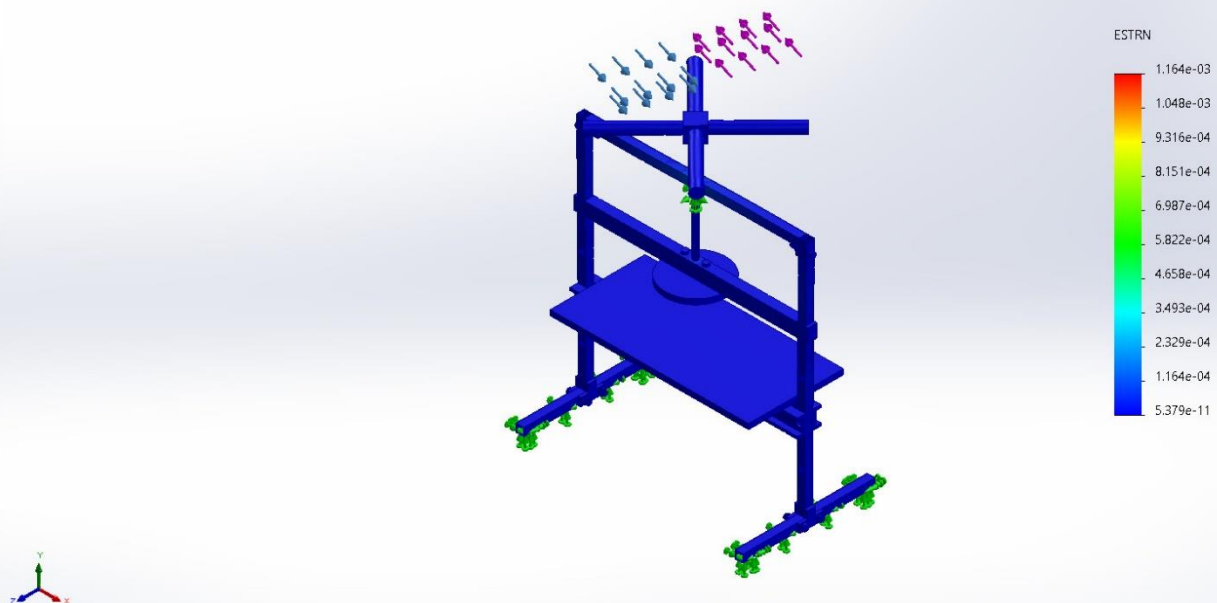
Model name: Screw Press
Study name: Static 1(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 24.6888



Screw Press-Static 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	5.379e-11 Element: 21230	1.164e-03 Element: 59603

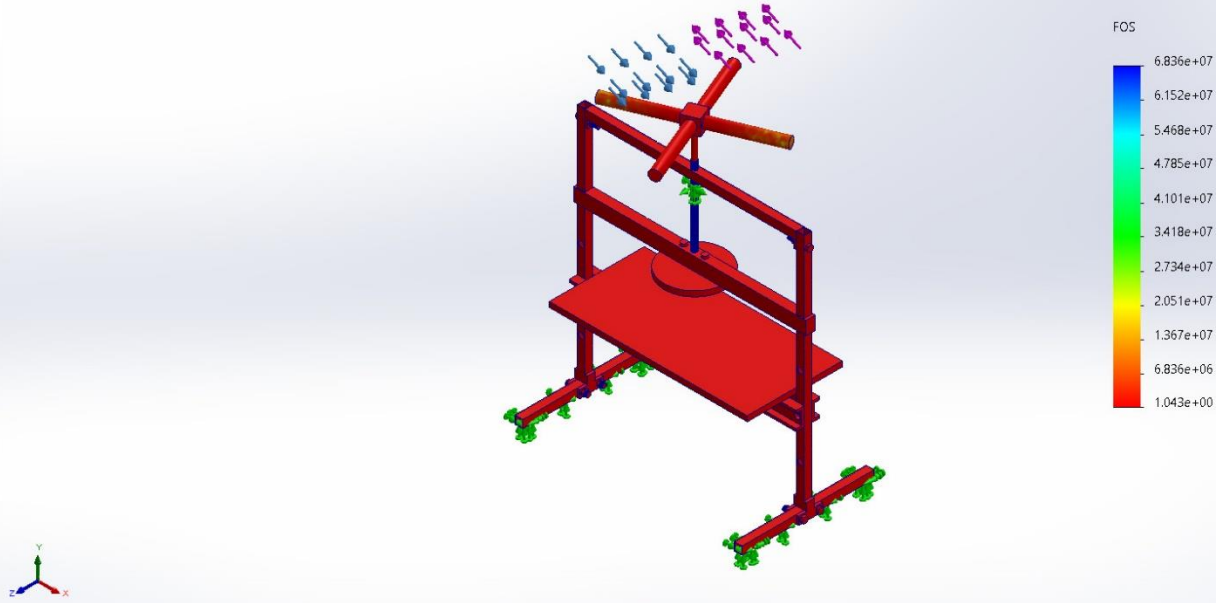
Model name: Screw Press
Study name: Static 1(-Default-)
Plot type: Static strain Strain1
Deformation scale: 24.6888



Screw Press-Static 1-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	1.043e+00 Node: 112657	6.836e+07 Node: 23841

Model name: Screw Press
Study name: Static 1(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Max von Mises Stress
Factor of safety distribution: Min FOS = 1



Screw Press-Static 1-Factor of Safety-Factor of Safety1

Conclusion

This static analysis of the screw press assembly was performed using SolidWorks Simulation to gain initial insights into its structural behavior under an applied load of 1000 N (two 500 N forces). The simulation aimed to assess stress distributions, displacements, and the resulting factor of safety.

The preliminary results indicate a maximum von Mises stress of 594.8 MPa and a minimum factor of safety of approximately 1.04. The maximum displacement observed was 4.78 mm. While these values provide a first look, several critical aspects of the current simulation setup necessitate a cautious interpretation and highlight areas for mandatory further refinement:

1. **Mesh Quality:** The mesh diagnostics revealed a maximum ratio significantly exceeding acceptable limits ($1.0974e+05$), with a considerable percentage of elements having aspect ratios greater than 10. This level of mesh distortion severely compromises the accuracy and reliability of stress, displacement, and factor of safety results. The current numerical values should not be considered definitive.
2. **Load Application and Magnitude:** The specific nature and intended representation of the applied 500 N forces need to be clearly correlated with the operational loads of the screw. This clarification is crucial for the relevance of the simulation. The total applied load of 1000 N should also be reconciled with the press's target force capacity outlined in the design specifications.
3. **Boundary Conditions:** The use of an edge fixture may introduce artificial stress concentrations. Reviewing and potentially revising this fixture to more accurately represent real-world constraints is recommended.

5- Summary of Results

Design analysis and human-powered screw press gave the following outputs:

Design Suitability:

A low-cost, compact design of a screw press was obtained, suitable for medium to small-sized workshops.

Customer requirements like low space, manpower, ease of operation, and heavy-duty were met in this design.

Mechanical Specifications:

The press can exert a maximum force of about 1.228 kN by a mechanism of man-powered power screw.

Handle diameter (250 mm) and screw pitch (3 mm) were selected for convenient operation and controlled motion.

750 mm × 360 mm bed area and 565 mm vertical daylight space offer numerous workshop applications.

Material and Construction Integrity:

Structural components were developed from high-strength steel alloys to ensure toughness under cyclic loading.

Simulation results indicated a maximum von Mises stress of 594.8 MPa and minimum factor of safety of 1.04, which would only make the design safe for the 1000 N load applied.

Simulation Analysis:

SolidWorks Static Analysis was employed in the analysis of stress distribution and deformation.

The maximum displacement was 4.78 mm, which is appropriate for the intended service of the press.

But mesh quality and boundary condition limits were observed and must be enhanced at subsequent stages of validation.

Manufacturing Readiness:

Assimilability and manufacturability have been considered in the design of the kit parts.

Detailed bill of material and CAD was submitted for prototyping.

Safety and Ergonomics:

The design has main safety features such as a wide stable base, clear shield, and minimized pinch points.

Ergonomic handle and handle height were included for the user's comfort while using it for an extended period.

6- Lessons Learned

throughout the process of design and analyzing, our team gained valuable skills in mechanical design and engineering development like

- 1-Importance of Customer-Centric Design
- 2- Value of Conceptual Planning
- 3-Material and Component Selection
- 4-Safety and Ergonomics
- 5- Simulation and Validation
- 6-Continuous Improvement
- 7- Documentation and Technical Communication.
- 8- FEA: finite element Analysis

